

Open Science Principles, Implementation, and Digital Transformation

Table Of Content

1. Introduction to Open Science and the Digital Turn
 - 1.1. Definition and Core Objectives of Open Science
 - 1.2. Open Science as a Consequence of Digital Transformation
2. Core Components and Infrastructure for Open Science
 - 2.1. Open Access to Publications, Data, Metadata, and Software
 - 2.2. Infrastructure Design Principles and Avoidance of Lock-in Effects

Table Of Content

- 3. Balancing Openness with Scientific Integrity and Governance
 - 3.1. Need for Protected Spaces for Discourse and Peer Review
 - 3.2. Risks of Surveillance and the Necessity of Balanced Incentives
- 4. Strategic Implementation and the Future of Research Culture
 - 4.1. DFG Fields of Activity and National/International Initiatives
 - 4.2. Conclusion on Values, Sovereignty, and Self-Governance

1. Introduction to Open Science and the Digital Turn

- DFG views Open Science as essential for digital transformation.
- Optimizes processes and boosts result re-usability.
- Openness must balance freedom, legal needs, and quality.
- Training is needed for new data and platform responsibilities.
- Open science requires careful governance, not just a "magic fix".

1.1. Definition and Core Objectives of Open Science

- **Goal:** Long-term access boosts researcher uptake.
- **Driver:** Digital transformation enables essential process functions.
- **Impact:** Optimizes workflows, stimulates discourse, builds trust.
- **Approach:** Shape frameworks while respecting community needs.

1.2. Open Science as a Consequence of Digital Transformation

- Open science is a necessary consequence, not just a buzzword.
- Optimize processes to boost re-usability for real impact.
- Transparency builds trust, but avoid treating it as a political panacea.
- Good research practice remains the bedrock; open science supports it.
- Balance openness with academic freedom and guard against lock-in.

2. Core Components and Infrastructure for Open Science

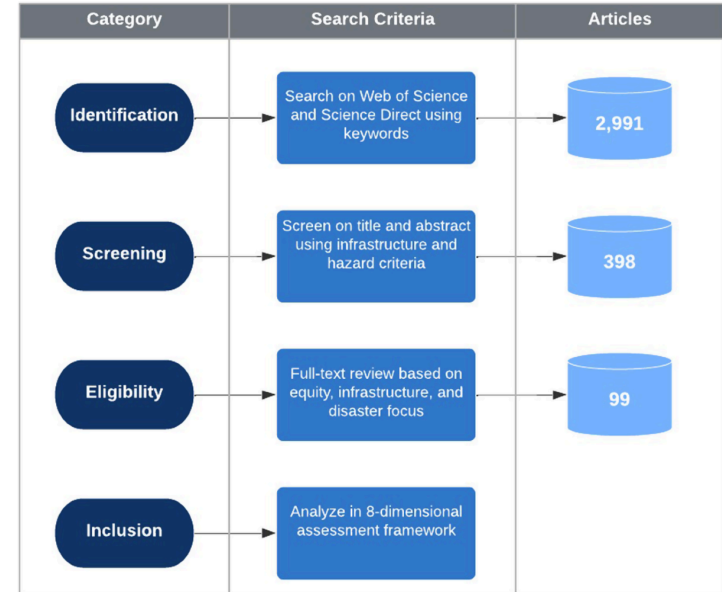
- DFG defines Open Science as long-term accessible practices.
- **Core:** Publications, Data, Software.
- **Access:** Ensures legally protected re-use of outputs.
- **Design:** Must avoid lock-in and proprietary dependencies.
- **Essentials:** Open code and metadata for reproducibility.

2.1. Open Access to Publications, Data, Metadata, and Software

- DFG funds open access and data/software infrastructures.
- Boosts transparency, replication, and cooperation.
- Society benefits from reproducible results and citizen science.
- Economy gains as industry data access improves innovation.
- Supports NFDI, EOSC, and the Diamond Open Access Plan.

2.2. Infrastructure Design Principles and Avoidance of Lock-in Effects

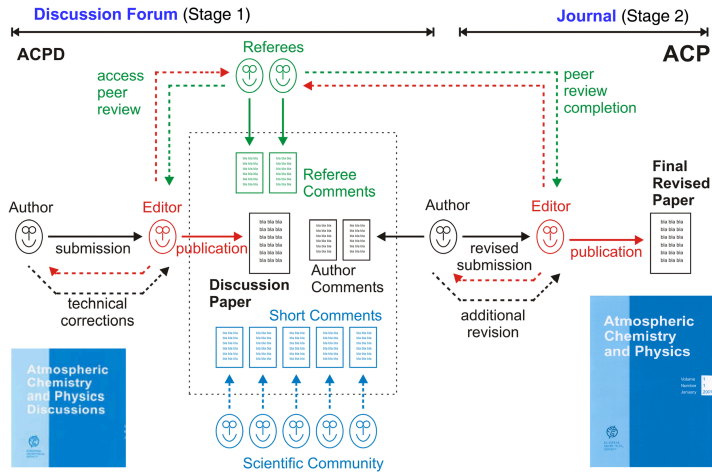
- Avoid lock-in effects to ensure infrastructural independence.
- Scientific bodies must govern infrastructure, not commercial providers.
- Preserve digital sovereignty to maintain research autonomy.
- Balance openness with legal limits and academic freedom.



3. Balancing Openness with Scientific Integrity and Governance

- Open science is a cultural shift, not just access.
- DFG sees it as a consequence of digital transformation.
- Optimize processes and boost result re-usability.
- Transparency strengthens trust, but not all data needs full openness.
- Balance freedom with necessary protections for discourse and IP.

3.1. Need for Protected Spaces for Discourse and Peer Review



- Why do we need protected spaces for discourse?
- Openness shouldn't silence critical scientific debates.
- Peer review needs safety from public pressure.
- Legal rights and IP can limit full transparency.
- Unregulated data access risks unwanted surveillance.

3.2. Risks of Surveillance and the Necessity of Balanced Incentives

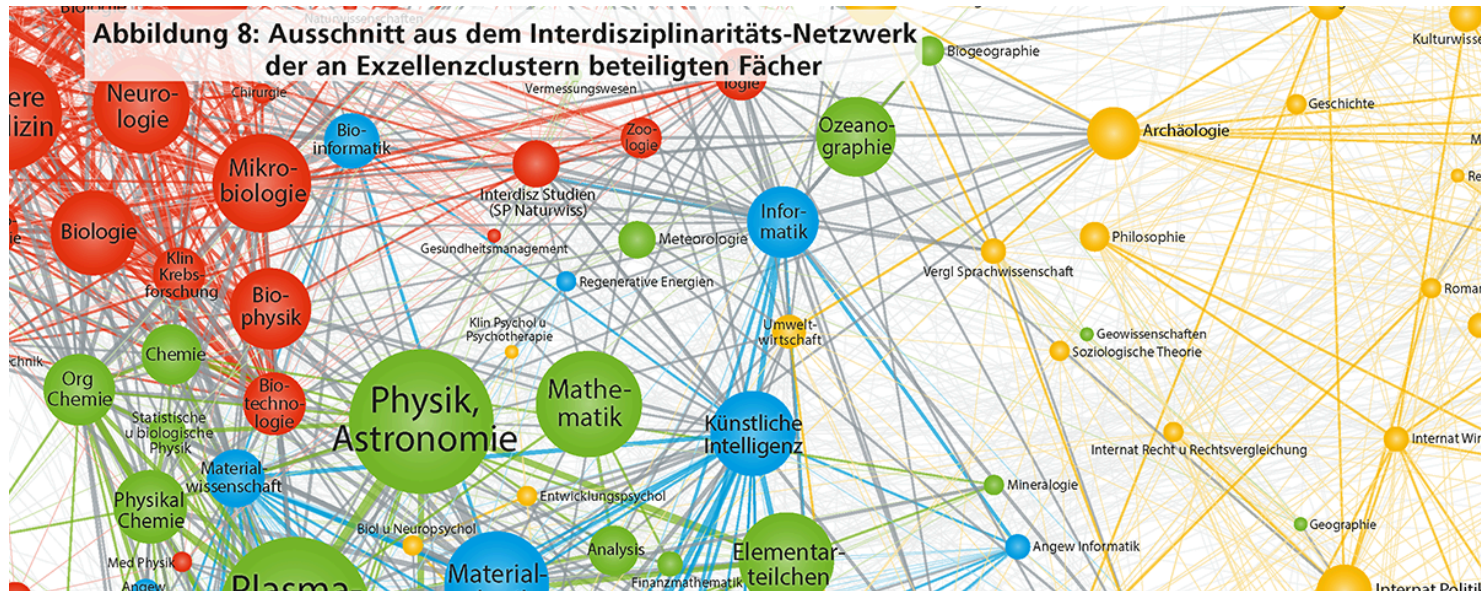
- Unregulated transparency risks unwanted surveillance.
- Commercial or state actors could exploit open data.
- Incentives must balance openness with scientific quality.
- Avoiding misuse protects academic integrity and trust.

4. Strategic Implementation and the Future of Research Culture

- Shape frameworks while respecting community needs.
- Open science optimizes processes and boosts reproducibility.
- Avoid treating openness as an ideology or end goal.
- Balanced incentives ensure quality over just societal impact.

4.1. DFG Fields of Activity and National/International Initiatives

- DFG funds open access & builds open science infrastructures.
- Active participation in national and international Open Science fora.
- Support 'Action Plan for Diamond Open Access' for science-led development.
- Contribute to NFDI and European Open Science Cloud (EOSC).



4.2. Conclusion on Values, Sovereignty, and Self-Governance

- Open science optimizes research but cannot replace core values like diligence.
- Preserve scientific autonomy to avoid lock-in effects and surveillance.
- Avoid treating open science as an end goal; it risks undermining quality.
- Self-governance ensures assessment focuses on reasoned quality.
- Unlock potential while addressing challenges without bias.

Thank you for listening!